

A simple method for logistic regression using time series data

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This version:

Suppose we are interested in forecasting daily rainfall in Gandhinagar on the basis of humidity levels (X) prevailing the previous day. Let Y denote the random variable which assumes a value of 1 if there is rain and 0 if there is no rain. Assuming that the successive values of Y are independent $p = E(Y)$ is the probability of rainfall on a particular day. This probability p is a function of X.

The method used for such problems to forecast the value of p at a future date is logistic regression.

It is assumed that $\log \frac{p_t}{1-p_t} = \beta_0 + \beta_1 X_{t-1}$, where $\frac{p_t}{1-p_t}$ denotes the odds in favor of rain. Thus $\frac{p_t}{1-p_t} = e^{\beta_0 + \beta_1 X_{t-1}}$ and so $p_t = \frac{e^{\beta_0 + \beta_1 X_{t-1}}}{1 + e^{\beta_0 + \beta_1 X_{t-1}}} = f(X_{t-1})$.

Let $\{y_1, \dots, y_T\}$ denote a sequence of observations of the random variable Y. The probability of obtaining this sequence of observations is given by $\prod_{t=1}^T \pi(t)$, where for $t \in \{1, \dots, T\}$, $\pi(t) = p_t^{y_t} (1 - p_t)^{(1-y_t)}$, and $p_t = f(X_{t-1})$ is the probability of rainfall on day t.

We use maximum likelihood estimation to estimate values of β_0 and β_1 , i.e. choose β_0 and β_1 to maximize the probability of the observed pattern of rainfalls the latter being equal to $\prod_{t=1}^T \pi(t)$.

This is one and the same as maximizing $\sum_{t=1}^T \log \pi(t) = \sum_{t=1}^T y_t \log p_t + \sum_{t=1}^T (1 - y_t) \log(1 - p_t) = \sum_{t=1}^T y_t [\log(e^{\beta_0 + \beta_1 X_{t-1}}) - \log(1 + e^{\beta_0 + \beta_1 X_{t-1}})] + \sum_{t=1}^T (1 - y_t) \log \frac{1}{1 + e^{\beta_0 + \beta_1 X_{t-1}}} = \sum_{t=1}^T y_t [\beta_0 + \beta_1 X_{t-1} - \log(1 + e^{\beta_0 + \beta_1 X_{t-1}})] - \sum_{t=1}^T (1 - y_t) \log(1 + e^{\beta_0 + \beta_1 X_{t-1}}) = \sum_{t=1}^T y_t [\beta_0 + \beta_1 X_{t-1}] - \sum_{t=1}^T \log(1 + e^{\beta_0 + \beta_1 X_{t-1}})$.

The above procedure is computationally complex and beyond the scope of Excel.

We therefore propose an alternative method.

For $t \in \{1, \dots, T\}$, let $\rho_t = \frac{1}{t} \sum_{\tau=1}^t y_\tau$ and $x_t = \frac{1}{t} \sum_{\tau=1}^t X_\tau$. Use these observed values to obtain the least squares estimate of β_0 and β_1 from the regression equation $\log \frac{\rho_t}{1-\rho_t} = \beta_0 + \beta_1 x_{t-1}$.

The method is not yet complete. We have only been able to solve the problem of obtaining T odds to run an OLS. Since the ρ_t 's are moving averages of the variety we have considered above,

if the model is correct then later values of ρ_t and x_t are expected to provide a more accurate fit, than what the earlier values would do. To rectify this situation, we model the sequence of residuals $\{e_1, \dots, e_T\}$ as follows: $e_t = \alpha e_{t-1} + \varepsilon$ for $t = 2, \dots, T$ where ε is a white noise. Here $e_t = \log \frac{\rho_t}{1-\rho_t} - \beta_0 - \beta_1 x_{t-1}$ for all $t = 1, \dots, T$. We estimate α by OLS using $e_0 = 0$ as initial condition.

Then we get $\log \frac{\rho_t}{1-\rho_t} = \beta_0 + \beta_1 x_{t-1} + \alpha e_{t-1} + \varepsilon$.

Thus the forecasted value of $\log \frac{p_t}{1-p_t} = \beta_0 + \beta_1 X_{t-1} + \alpha e_{t-1}$.

Note that the forecasted value is that of $\log \frac{p_t}{1-p_t}$, which is the odds ratio at time t , instead of $\log \frac{\rho_t}{1-\rho_t}$. Further instead of x_{t-1} (average humidity level over the first $t-1$ time periods) we use X_{t-1} (humidity level at $t-1$) on the right hand side.

This is easily solvable by EXCEL.

For details on concepts discussed here and for background material one may refer to Shmueli, Patel and Bruce (2007).

References

1. Shmueli, G., N.R. Patel and P.C. Bruce (2007): “Data Mining for Business Intelligence: Concepts, Techniques, and Applications in Microsoft Office Excel with XLMiner”, John Wiley & Sons, Inc.

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